Lab 10: Fixed Effects, Random Effects, and Difference-in-Differences

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1 Goals

Parts 1 and 2 follow the lecture through the Kleven et al. slides, on the same marriage data. Every number below should match a slide.

1. Show that in a 2×2, the difference in differences of four means, the interaction regression, two-way fixed effects (TWFE), and first differences are **the same number**.
2. Estimate TWFE with `feols()` (and `felm()`), three ways to the same number, and cluster the standard errors by man.
3. Check strict exogeneity with a lead.
4. Estimate random effects, compare it with fixed effects, and **justify** the choice with the Hausman test.

Part 3 (staggered adoption) goes with the slides after Kleven et al.

```
## install.packages(c("wooldridge", "plm", "fixest", "lfe", "stargazer")) once,
if needed
library(dplyr)
library(ggplot2)
library(fixest)     ## feols(): fixed effects, clustered standard errors (our
default)
library(lfe)        ## felm(): the same, with fixed effects, IV, and clusters
in one formula
library(stargazer)  ## formatted regression tables (reads felm, lm, plm; not
feols)
library(plm)        ## plm(): random effects, Hausman test
library(wooldridge) ## the wagepan data

## plm also has functions called lag() and lead(); we use dplyr's
lag <- dplyr::lag; lead <- dplyr::lead
```

2 Part 1: The 2×2

2.1 The data

```
data(wagepan)
w <- wagepan
## one row per man per year: 545 men x 8 years (1980-87) = 4,360 rows
c(men = n_distinct(w$nr), years = n_distinct(w$year), rows = nrow(w))
#>   men years  rows
#>   545     8  4360
## the variables we use: nr = man id, lwage = log hourly wage,
## married = 1 if married that year
head(w[, c("nr", "year", "lwage", "married")], 3)
```

```
#>   nr year lwage married
#> 1 13 1980 1.198       0
#> 2 13 1981 1.853       0
#> 3 13 1982 1.344       0
```

The panel is **balanced**: every man appears in every year. That is what lets double demeaning equal the dummies later on.

2.2 Build the 2×2 sample

As in the lecture: 1980 and 1987 only, and the men who married in between or did not change (the 17 who left marriage are set aside).

```
## each man's status and wage in 1980 and in 1987, side by side
d80 <- w |> filter(year == 1980) |> select(nr, y80 = lwage, D80 = married)
d87 <- w |> filter(year == 1987) |> select(nr, y87 = lwage, D87 = married)
wide <- inner_join(d80, d87, by = "nr") |>
  mutate(dD = D87 - D80,       ## change in marital status: -1, 0, or 1
         dy = y87 - y80)       ## wage growth, 1980 to 1987
table(wide$dD)
#>
#>  -1   0   1
#>  17 277 251

## keep the men who married (dD = 1) or did not change (dD = 0)
keep <- wide |> filter(dD %in% c(0, 1))
nrow(keep)
#> [1] 528

## back to long form: two rows per man; treated = married in between
long22 <- w |>
  filter(year %in% c(1980, 1987), nr %in% keep$nr) |>
  mutate(treated = as.integer(nr %in% keep$nr[keep$dD == 1]),
         post    = as.integer(year == 1987))
nrow(long22)
#> [1] 1056
```

2.3 Pooled OLS

```
## treats the 1,056 person-years as 1,056 different men
round(coeftable(feols(lwage ~ married, long22, cluster = ~nr)), 3)
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept)    1.488      0.023   63.85        0
```

```
#> married        0.358      0.031   11.42        0
#> attr(,"vcov_type")
#> [1] "Clustered (nr)"
```

0.358: the slide's pooled estimate. It mixes the marriage effect with everything stable that differs between men who marry and men who do not.

2.4 Four routes to one number

```
## Route 1: difference in differences of the four cell means
cm <- long22 |> group_by(treated, post) |> summarise(m = mean(lwage), .groups =
"drop")
cm
#> # A tibble: 4 x 3
#>   treated  post     m
#>     <int> <int> <dbl>
#> 1       0     0  1.35
#> 2       0     1  1.82
#> 3       1     0  1.43
#> 4       1     1  1.92
m <- function(g, p) cm$m[cm$treated == g & cm$post == p]
did_hand <- (m(1, 1) - m(1, 0)) - (m(0, 1) - m(0, 0))

## Route 2: the interaction regression, Y = a + g*treated + d*post +
tau*treated*post
did_int <- coef(lm(lwage ~ treated * post, long22))[["treated:post"]]

## Route 3: TWFE, a dummy for each man and each year
twfe22 <- coef(feols(lwage ~ married | nr + year, long22))[["married"]]

## Route 4: first differences, wage growth on change in status, with an
intercept
fd22 <- coef(lm(dy ~ dD, keep))[["dD"]]

round(c(DiD_by_hand = did_hand, interaction = did_int,
        TWFE = twfe22, first_diff = fd22), 5)
#> DiD_by_hand interaction        TWFE  first_diff
#>      0.0125      0.0125      0.0125      0.0125
```

All four are 0.0125, the slide's 0.012. Why they must agree:

- In the interaction regression, `treated` plays the role of the man dummies (it absorbs the level difference between the two groups) and `post` plays the role of the year dummy.

- With two years, subtracting each man's mean is the same as taking his change (his two demeaned values are $-\Delta/2$ and $+\Delta/2$), so TWFE is first differences.
- First differences with an intercept and a 0/1 regressor is a difference in means: married men's growth minus the others' growth.

```
## the last point, directly: mean growth by group
keep |> group_by(dD) |> summarise(men = n(), mean_growth = mean(dy))
#> # A tibble: 2 x 3
#>      dD   men mean_growth
#>   <int> <int>       <dbl>
#> 1     0   277       0.476
#> 2     1   251       0.488
```

$0.488 - 0.476 = 0.012$.

3 Part 2: All Men, All Years

3.1 Pooled, man FE, year FE, two-way FE

`feols()` is our default. The part after `|` lists the fixed effects, and `cluster = ~nr` clusters the standard errors by man.

```
## all 545 men (including the 17 who left marriage), all eight years
pool   <- feols(lwage ~ married,             w, cluster = ~nr)
fe_man <- feols(lwage ~ married | nr,        w, cluster = ~nr)  ## man FE only
fe_yr  <- feols(lwage ~ married | year,      w, cluster = ~nr)  ## year FE
only
twfe   <- feols(lwage ~ married | nr + year, w, cluster = ~nr)  ## both: TWFE
round(c(pooled = coef(pool)[["married"]], man_FE = coef(fe_man)[["married"]],
        year_FE = coef(fe_yr)[["married"]], TWFE = coef(twfe)[["married"]]), 3)
#>  pooled  man_FE year_FE    TWFE
#>   0.220   0.243   0.150   0.061
```

The slides' four numbers: 0.220, 0.243, 0.150, 0.061. `stargazer` cannot read `feols` objects, so for a formatted table we fit the same four models with `felm()` from the `lfe` package. Its formula has four parts separated by `|`: `y ~ x | fixed effects | instruments | cluster` (0 means none).

```
p_l  <- felm(lwage ~ married | 0         | 0 | nr, data = w)
m_l  <- felm(lwage ~ married | nr        | 0 | nr, data = w)
y_l  <- felm(lwage ~ married | year      | 0 | nr, data = w)
tw_l <- felm(lwage ~ married | nr + year | 0 | nr, data = w)
stargazer(p_l, m_l, y_l, tw_l, type = "latex", header = FALSE, float = FALSE,
          keep = "married", keep.stat = "n",
          column.labels = c("pooled", "man FE", "year FE", "two-way FE"),
```

```
          add.lines = list(c("Man FE", "", "Yes", "", "Yes"),
                           c("Year FE", "", "", "Yes", "Yes")),
          notes = "Standard errors clustered by man.")
```

	Dependent variable:			
	lwage			
	pooled	man FE	year FE	two-way FE
	(1)	(2)	(3)	(4)
married	0.220***	0.243***	0.150***	0.061***
	(0.026)	(0.022)	(0.028)	(0.021)
Man FE		Yes		Yes
Year FE			Yes	Yes
Observations	4,360	4,360	4,360	4,360
Note:				*p<0.1; **p<0.05; ***p<0.01
				Standard errors clustered by man.

3.2 Three routes to the TWFE estimate

```
## Route 1: dummies. lm() drops one year automatically (collinearity)
b_dummies <- coef(lm(lwage ~ married + factor(nr) + factor(year),
w))[["married"]]

## Route 2: double demeaning by hand
w <- w |>
  group_by(nr)   |> mutate(y_man = mean(lwage), D_man = mean(married)) |>
  group_by(year) |> mutate(y_yr  = mean(lwage), D_yr  = mean(married)) |>
  ungroup() |>
  mutate(y_dd = lwage   - y_man - y_yr + mean(lwage),     ## minus man mean,
  minus year
         D_dd = married - D_man - D_yr + mean(married))   ## mean, plus grand
         mean
## no intercept: every double-demeaned variable has mean zero
b_demeaned <- coef(lm(y_dd ~ 0 + D_dd, w))[["D_dd"]]

## Route 3: feols
b_fixest <- coef(twfe)[["married"]]

round(c(dummies = b_dummies, double_demeaned = b_demeaned, feols = b_fixest),
6)
```

```
#>         dummies double_demeaned           feols
#>         0.06058         0.06058         0.06058
```

Identical to six decimals: two-way fixed effects **is** a comparison of each man with himself, net of what happened to every man that year.

3.3 How to cluster by man

```
## feols: add cluster = ~nr (the default in this lab); vcov = "iid" turns it
off
tw_iid <- feols(lwage ~ married | nr + year, w, vcov = "iid")
## felm: put the cluster variable in the fourth part of the formula
tw_felm <- felm(lwage ~ married | nr + year | 0 | nr, data = w)
round(c(feols_iid = se(tw_iid)[["married"]],
        feols_cluster = se(twfe)[["married"]],
        felm_cluster = tw_felm$cse[["married"]]), 4)
#>     feols_iid feols_cluster  felm_cluster
#>        0.0184        0.0215        0.0215
```

Same coefficient; the clustered standard error is larger (0.0215 vs 0.0184). `feols` and `felm` agree up to a small-sample correction in the fourth decimal. A man's wage shocks are correlated across his eight years, so eight years of one man are worth less than eight independent observations: always cluster by the unit.

3.4 Strict exogeneity: add the lead

```
## f(married) = next year's married; needs panel.id so feols knows the order.
## 1987 rows are dropped (no next year).
lead_fit <- feols(lwage ~ married + f(married) + union + expersq | nr + year,
                  w, panel.id = ~nr + year, cluster = ~nr)
round(coeftable(lead_fit), 4)
#>            Estimate Std. Error t value Pr(>|t|)
#> married      0.0477     0.0247  1.9329   0.0538
#> f(married)  -0.0005     0.0221 -0.0232   0.9815
#> union        0.0803     0.0240  3.3421   0.0009
#> expersq     -0.0055     0.0010 -5.5648   0.0000
#> attr(,"vcov_type")
#> [1] "Clustered (nr)"
```

Next year's marriage does not predict this year's wage ($p = 0.98$): no sign of feedback. But its standard error (0.022) is as large as the effect itself.

3.5 Random effects

```
## plm needs the panel structure declared: id first, then time
pw <- pdata.frame(as.data.frame(wagepan), index = c("nr", "year"))

## the slides' specification: married, union, experience, education, race,
years
re <- plm(lwage ~ married + union + educ + black + hisp + exper + expersq +
factor(year),
          data = pw, model = "random")
round(coef(re)[c("married", "union", "educ")], 4)
#> married   union    educ
#>  0.0640  0.1061  0.0919

## the variance components and theta
ercomp(re)
#>                  var std.dev share
#> idiosyncratic 0.123   0.351  0.54
#> individual    0.105   0.325  0.46
#> theta: 0.643
```

Random effects can estimate `educ` (fixed over time); fixed effects cannot. Check θ by hand from the two variances:

```
s2_e <- ercomp(re)$sigma2[["idios"]]   ## variance of epsilon (within)
s2_a <- ercomp(re)$sigma2[["id"]]      ## variance of the man effect alpha
(between)
c(theta = 1 - sqrt(s2_e / (s2_e + 8 * s2_a)))   ## T = 8 years per man
#>  theta
#> 0.6429
```

Random effects subtracts 64% of each man's mean; fixed effects subtracts all of it; pooled OLS, none.

3.6 Comparing: pooled, RE, FE

```
## pooled and FE with felm (clustered by man), RE with plm
pool_c <- felm(lwage ~ married + union + educ + black + hisp + exper + expersq
| year | 0 | nr,
               data = w)
fe_c   <- felm(lwage ~ married + union + expersq | nr + year | 0 | nr, data =
w)
stargazer(pool_c, re, fe_c, type = "latex", header = FALSE, float = FALSE,
```

```
          keep = c("married", "union", "educ"), keep.stat = "n",
          column.labels = c("pooled", "random effects", "fixed effects"),
          model.names = FALSE,
          notes = "Pooled and FE: clustered by man. All include year effects.")
```

	Dependent variable:		
	lwage		
	pooled	random effects	fixed effects
	(1)	(2)	(3)
married	0.108***	0.064***	0.047**
	(0.026)	(0.017)	(0.021)
union	0.182***	0.106***	0.080***
	(0.027)	(0.018)	(0.023)
educ	0.091***	0.092***	
	(0.011)	(0.011)	
Observations	4,360	4,360	4,360

Note: *p<0.1; **p<0.05; ***p<0.01

Pooled and FE: clustered by man. All include year effects.

0.108, 0.064, 0.047. Random effects sits between pooled and fixed: it partly uses the comparison between men. Only pooled and RE can estimate education.

3.7 Justifying the choice: Hausman

```
## the same regressors in both (FE cannot estimate educ or race)
fe_plm <- plm(lwage ~ married + union + expersq + factor(year), pw, model =
"within")
re_plm <- plm(lwage ~ married + union + expersq + factor(year), pw, model =
"random")
phtest(fe_plm, re_plm)
#>
#>  Hausman Test
#>
#> data:  lwage ~ married + union + expersq + factor(year)
#> chisq = 37, df = 10, p-value = 0.00006
#> alternative hypothesis: one model is inconsistent
```

$\chi^2 = 37.0$: the FE and RE coefficients differ by more than noise, so the random-effects assumption (man effects unrelated to the regressors) is rejected. One caveat to report with it: the test assumes random effects is fully efficient, which fails when errors are serially correlated.

> **Note**
>
> **How to write the justification.** "We estimate the marriage premium with man and year fixed effects, so stable differences between men who marry and men who do not cannot bias the estimate. A random-effects model would also use comparisons between men; a Hausman test rejects its assumption that the man effects are unrelated to the regressors ($\chi^2_{10} = 37.0$, $p < 0.001$), and the married coefficient falls from 0.064 (RE) to 0.047 (FE). We therefore report fixed effects, with standard errors clustered by man."

4 Part 3: Staggered Adoption (after the Kleven et al. slides)

This part goes with the next section of the lecture.

```
library(did)
library(bacondecomp)

data(base_stagg, package = "fixest")
stg <- base_stagg |> as.data.frame() |>
  mutate(D = as.integer(time_to_treatment >= 0),
         cohort = year_treated) |>            # 10000 codes never-treated
  select(id, year, y, D, cohort, time_to_treatment, x1, treatment_effect_true)

c(units = n_distinct(stg$id), periods = n_distinct(stg$year),
  cohorts = n_distinct(stg$cohort),
  never_treated_units = sum(!duplicated(stg$id) & stg$cohort == 10000))
#>               units             periods             cohorts
#>                  95                  10                  10
#> never_treated_units
#>                  50
```

The simulated panel has 100 units over 10 periods. Half are never treated; the other half adopt in staggered fashion, one cohort per period.

4.1 The staggered problem

`base_stagg` is simulated, so we know the answer. The column `treatment_effect_true` records each observation's true effect.

```
true_att <- mean(stg$treatment_effect_true[stg$D == 1])
c(true_ATT = true_att)
```

```
#> true_ATT
#>       -1
```

```
stg |> filter(D == 1) |>
  group_by(time_to_treatment) |>
  summarise(truth = mean(treatment_effect_true)) |>
  ggplot(aes(time_to_treatment, truth)) +
  geom_line(colour = "navy", linewidth = .9) + geom_point(colour = "navy") +
  labs(x = "periods since treatment", y = "true effect") +
  theme_minimal(base_size = 10)
```

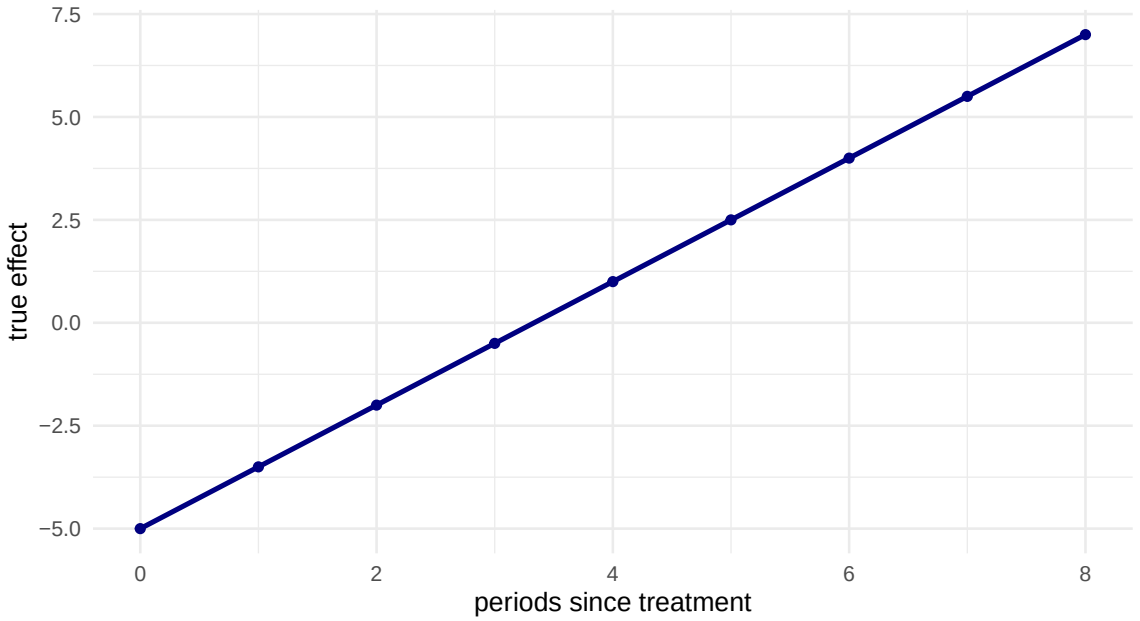


Figure 1: The true effect by time since treatment. It grows in magnitude with exposure — exactly the dynamic pattern that breaks TWFE.

4.1.1 What TWFE reports

```
m_twfe <- feols(y ~ D | id + year, data = stg, cluster = ~ id)
c(TWFE = coef(m_twfe)["D"], truth = true_att)
#> TWFE.D  truth
#> -3.468 -1.000
```

> **!** Important
>
> The true ATT is −1. TWFE reports roughly −3.5 — **three and a half times too large**. Nothing is misspecified in the usual sense: there is no omitted variable, no functional-form error, no measurement problem. The failure comes entirely from using already-treated units

as controls while their own effects are still growing.

4.1.2 Where the damage comes from

```
bd <- bacon(y ~ D, data = stg, id_var = "id", time_var = "year")
#>                        type weight avg_est
#> 1 Earlier vs Later Treated 0.1429  -2.460
#> 2 Later vs Earlier Treated 0.1429  -8.485
#> 3     Treated vs Untreated 0.7143  -2.666

bd |> group_by(type) |>
  summarise(total_weight = sum(weight),
            weighted_estimate = sum(estimate * weight) / sum(weight),
            n_comparisons = n(), .groups = "drop") |>
  knitr::kable(digits = 4)
```

type	total_weight	weighted_estimate	n_comparisons
Earlier vs Later Treated	0.1429	-2.460	36
Later vs Earlier Treated	0.1429	-8.485	36
Treated vs Untreated	0.7143	-2.666	9

Read the `Earlier vs Later Treated` and `Later vs Earlier Treated` rows. The second is the **forbidden comparison** — already-treated units serving as controls. Its weight tells you how much of the TWFE estimate is contaminated.

```
ggplot(bd, aes(weight, estimate, colour = type, shape = type)) +
  geom_point(size = 2, alpha = .8) +
  geom_hline(yintercept = coef(m_twfe)["D"], linetype = 2) +
  geom_hline(yintercept = true_att, colour = "firebrick") +
  labs(x = "weight", y = "2x2 DID estimate", colour = NULL, shape = NULL) +
  theme_minimal(base_size = 9) + theme(legend.position = "bottom")
```

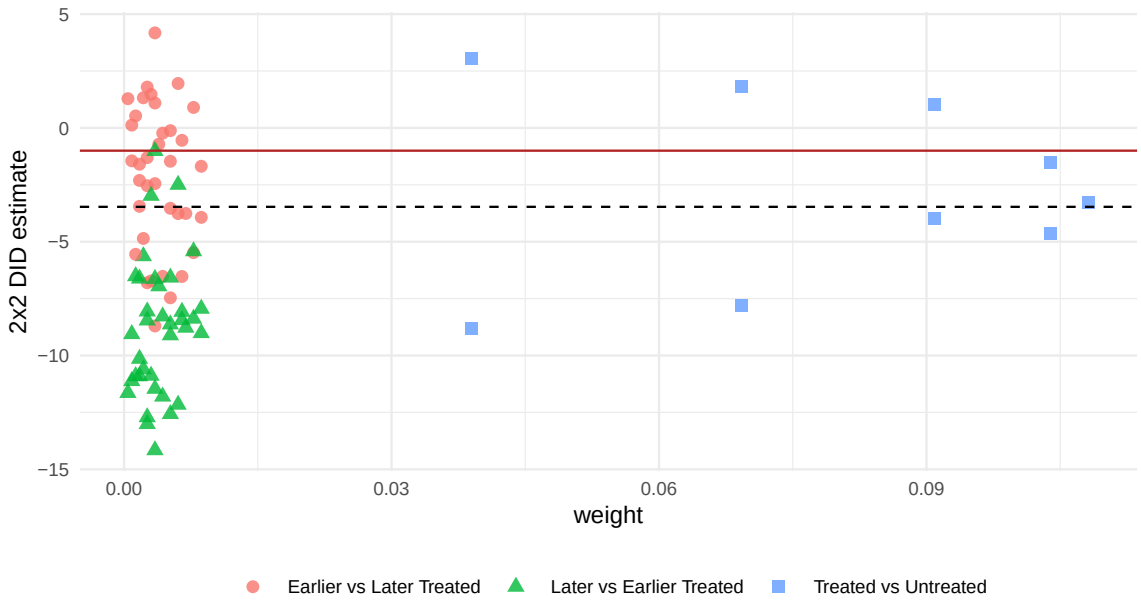


Figure 2: Goodman-Bacon decomposition: each point is a 2x2 comparison, positioned by its weight and its estimate.

The dashed line is the TWFE estimate — a weighted average of everything shown. The red line is the truth.

4.2 Estimators that work

4.2.1 Sun–Abraham

```
m_sa <- feols(y ~ sunab(cohort, year) | id + year,
              data = stg, cluster = ~ id)
summary(m_sa, agg = "att")
#> OLS estimation, Dep. Var.: y
#> Observations: 950
#> Fixed-effects: id: 95,  year: 10
#> Standard-errors: Clustered (id)
#>     Estimate Std. Error t value Pr(>|t|)
#> ATT  -0.7552     0.4211  -1.793 0.076144 .
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#> RMSE: 1.98883     Adj. R2: 0.479262
#>                 Within R2: 0.424684
```

```
iplot(m_sa, main = "", xlab = "periods since treatment")
```

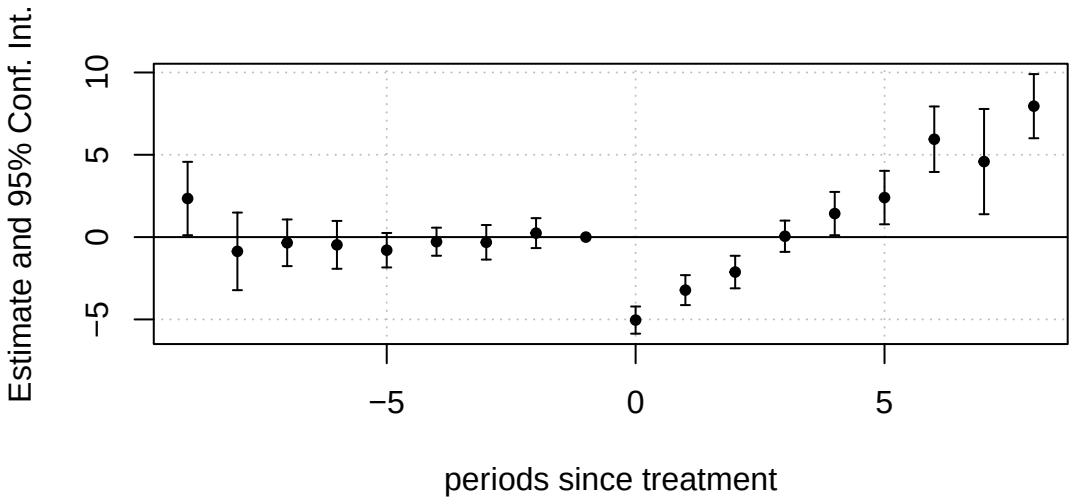


Figure 3: Sun–Abraham event study. Pre-period coefficients hover around zero; post-period coefficients trace the true dynamic path.

4.2.2 Callaway–Sant'Anna

```
set.seed(723)
cs <- att_gt(yname = "y", tname = "year", idname = "id",
             gname = "cohort", data = stg,
             control_group = "notyettreated", bstrap = TRUE, cband = FALSE)
cs_att <- aggte(cs, type = "simple", na.rm = TRUE)
summary(cs_att)
#>
#> Call:
#> aggte(MP = cs, type = "simple", na.rm = TRUE)
#>
#> Reference: Callaway, Brantly and Pedro H.C. Sant'Anna.
"Difference-in-Differences with Multiple Time Periods." Journal of
Econometrics, Vol. 225, No. 2, pp. 200-230, 2021.
<https://doi.org/10.1016/j.jeconom.2020.12.001>,
<https://arxiv.org/abs/1803.09015>
#>
#>
#>      ATT    Std. Error     [ 95%  Conf. Int.]
#>  -0.7995        0.7115     -2.194      0.5949
#>
#>
#> ---
```

```
#> Signif. codes: `*' confidence band does not cover 0
#>
#> Control Group:  Not Yet Treated,  Anticipation Periods:  0
#> Estimation Method:  Doubly Robust
```

```
cs_dyn <- aggte(cs, type = "dynamic", na.rm = TRUE)
ggdid(cs_dyn) + theme_minimal(base_size = 9)
```

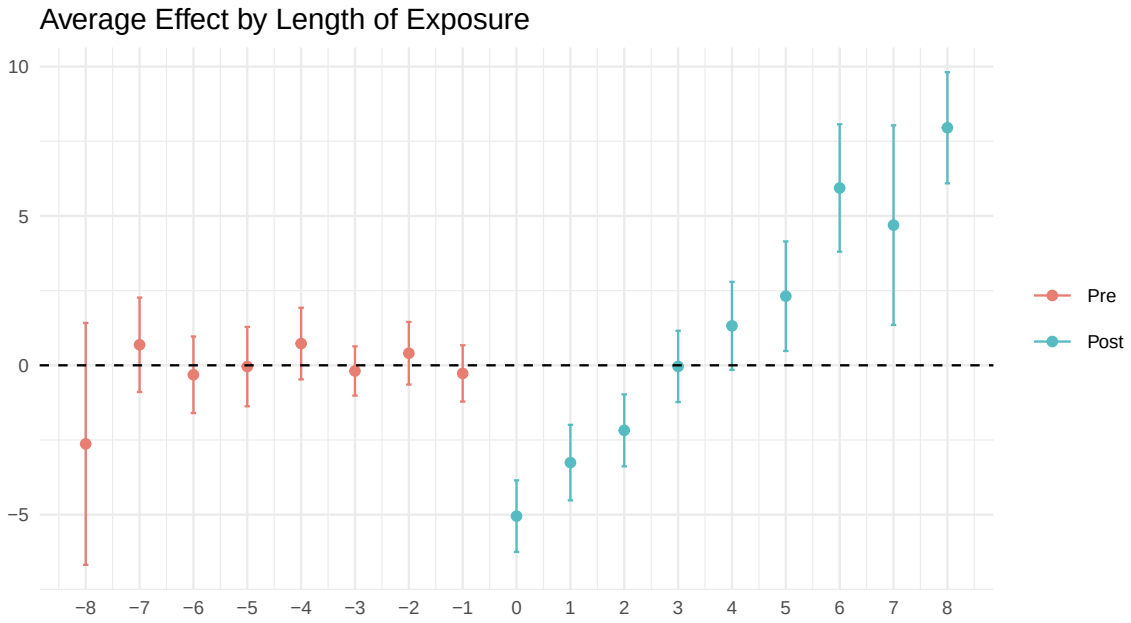


Figure 4: Callaway–Sant'Anna dynamic effects, aggregated across cohorts by time since treatment.

4.2.3 The scoreboard

```
tibble::tibble(
  estimator = c("Truth", "TWFE", "Sun-Abraham", "Callaway-Sant'Anna"),
  estimate = c(true_att, coef(m_twfe)["D"],
               summary(m_sa, agg = "att")$coeftable[1, 1],
               cs_att$overall.att)
) |> knitr::kable(digits = 4)
```

estimator	estimate
Truth	-1.0000
TWFE	-3.4676
Sun-Abraham	-0.7552
Callaway-Sant'Anna	-0.7995

Sun–Abraham and Callaway–Sant'Anna land near the truth of −1. TWFE does not, and TWFE is the specification that appears in thousands of published papers.

(The two modern estimators are not exactly −1 because the "simple" aggregation weights cohorts and periods differently from a naive average over treated observations. That is a choice of estimand, not an error — see Exercise 3.)

4.3 A real application

`did::mpdta` contains county-level teen employment and the timing of minimum-wage increases, 2003–2007.

```
data(mpdta, package = "did")
c(counties = n_distinct(mpdta$countyreal), years = n_distinct(mpdta$year),
  cohorts = paste(sort(unique(mpdta$first.treat)), collapse = ", "))
#>              counties                 years               cohorts
#>                 "500"                   "5" "0, 2004, 2006, 2007"
```

```
mpdta |>
  group_by(first.treat, year) |>
  summarise(y = mean(lemp), .groups = "drop") |>
  ggplot(aes(year, y, colour = factor(first.treat))) +
  geom_line(linewidth = .8) + geom_point() +
  labs(x = NULL, y = "mean log teen employment", colour = "cohort") +
  theme_minimal(base_size = 10)
```

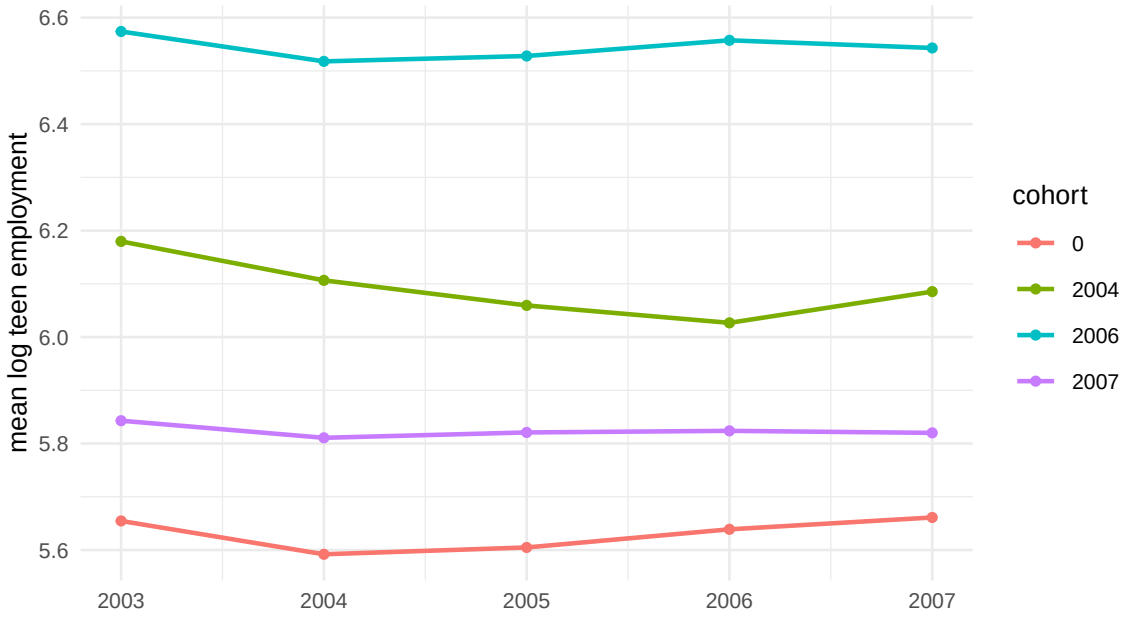


Figure 5: Raw group means of log teen employment by treatment cohort. Always draw this before any regression.

```r
## `treat` in mpdta is a time-invariant cohort flag; build the time-varying
## treatment indicator ourselves
mp <- mpdta |> mutate(D = as.integer(first.treat > 0 & year >= first.treat))

m_twfe_mp <- feols(lemp ~ D | countyreal + year, data = mp,
                   cluster = ~ countyreal)

set.seed(723)
cs_mp <- att_gt(yname = "lemp", tname = "year", idname = "countyreal",
                gname = "first.treat", xformla = ~ lpop, data = mp,
                control_group = "notyettreated", bstrap = TRUE)
cs_mp_att <- aggte(cs_mp, type = "simple", na.rm = TRUE)

c(TWFE = coef(m_twfe_mp)["D"], CS = cs_mp_att$overall.att)
#>   TWFE.D       CS
#> -0.03655 -0.04135
```

```r
ggdid(aggte(cs_mp, type = "dynamic", na.rm = TRUE)) + theme_minimal(base_size =
9)
```

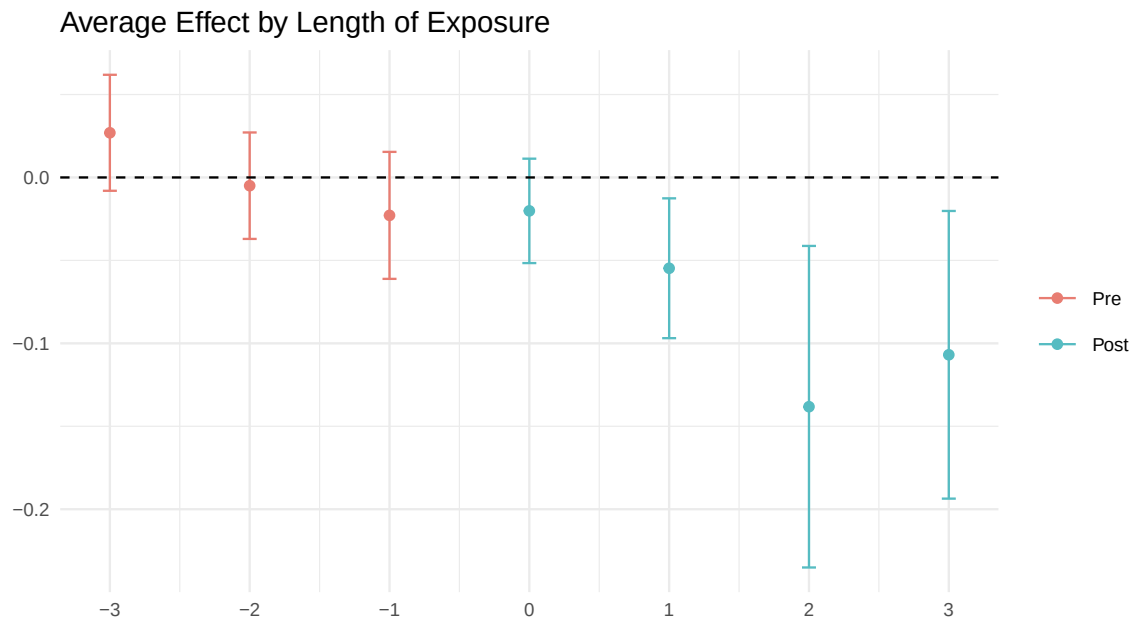


Figure 6: Event-study estimates for the minimum-wage application, with conditional parallel trends adjusting for county population.

Note the `xformla = ~ lpop` argument: this is **conditional** parallel trends, with the adjustment done by a doubly robust estimator. Week 6 has reappeared inside a DID.

4.4 How bad could parallel trends be?

Rather than testing pre-trends and proceeding, bound the violation.

```
## A crude linear-extrapolation diagnostic -- NOT a Rambachan-Roth bound.
## It ignores sampling uncertainty and assumes the violation stays linear.
dyn <- aggte(cs_mp, type = "dynamic", na.rm = TRUE)
pre <- data.frame(e = dyn$egt, att = dyn$att.egt) |> filter(e < 0)
post_mean <- mean(dyn$att.egt[dyn$egt >= 0])

## slope of the estimated pre-trend
pre_slope <- if (nrow(pre) > 1) coef(lm(att ~ e, data = pre))["e"] else NA
c(pre_trend_slope = pre_slope, mean_post_effect = post_mean,
  periods_to_explain = post_mean / pre_slope)
#>    pre_trend_slope.e     mean_post_effect periods_to_explain.e
#>              -0.02490             -0.07999              3.21299
```

Interpretation: if the untreated potential outcome had continued drifting at the estimated pre-trend slope, how many periods of drift would it take to account for the whole post-treatment effect? A small number means the design is fragile.

> **Warning**
>
> This is a back-of-envelope diagnostic, not a formal sensitivity analysis. It treats the estimated pre-trend as if it were known, ignores uncertainty in both pieces, and assumes any violation is linear. For the real thing, use `HonestDiD` and report the **breakdown value** $\bar{M}$, how large the post-treatment violation may be, relative to the largest violation seen before treatment, before your conclusion reverses.

5 Exercises

Parts 1 and 2

1. **The 2×2 with the 17 men who left marriage.** Put them back (`dD` in $\{-1, 0, 1\}$). The interaction regression needs a 0/1 `treated`, so run first differences with an intercept instead. You should get 0.051. Explain in one sentence why it is not $0.488 - 0.476$.

2. **Two packages, one answer.** Re-estimate the pooled-RE-FE comparison's fixed-effects column with `feols(... | nr + year, cluster = ~nr)` and with `felm(... | nr + year | 0 | nr)`. Confirm the coefficients agree, and say where the standard errors differ and why.

3. **Random trends.** Estimate `feols(lwage ~ married | nr[exper] + year, w, cluster = ~nr)`: each man gets his own experience slope. Compare with TWFE and say what the difference means.

4. **Write the justification for your own data.** For a panel you use, say which unit-level traits could be related to the treatment, and whether you would report fixed or random effects. Name the test you would run.

Part 3

5. **Build the failure yourself.** Simulate a staggered design with three cohorts and a *constant* treatment effect of 2. Show that TWFE recovers it. Then make the effect grow with exposure and show that TWFE does not. Explain the difference in one sentence.

6. **Not-yet-treated versus never-treated.** Re-run `att_gt()` with `control_group = "nevertreated"`. Compare with the not-yet-treated version. Which is more efficient, and under what assumption is each valid?

7. **Aggregation choices.** From the same `att_gt` object, compute `aggte(type = "group")` and `aggte(type = "calendar")`. These are three different estimands built from the same group-time effects. Write one sentence describing each in words.

8. **Pre-trend power.** For the `mpdta` application, simulate what a linear pre-trend of a given slope would do to the estimated pre-period coefficients. How large would the slope have to be before your pre-trend test detects it 80% of the time?

9. **Levels or logs?** `mpdta` supplies only log employment, so construct `emp <- exp(lemp)` and re-run the minimum-wage analysis in levels. Do the conclusions change? Which scale would you have pre-committed to, and why? (Recall from lecture that parallel trends in levels and in logs are different , and generally incompatible, assumptions.)

10. **Clustering.** Re-estimate the `mpdta` TWFE model clustering on state (`countyreal %/% 1000` gives the state FIPS code) rather than county. How many clusters are there, and what does Week 2 say you should do about it?

6 Session information

```
sessionInfo()
#> R version 4.5.2 (2025-10-31)
#> Platform: aarch64-apple-darwin20
#> Running under: macOS Tahoe 26.6.2
#>
#> Matrix products: default
#> BLAS:
/System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versio

#> LAPACK:
/Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
LAPACK version 3.12.1
#>
```

```
#> locale:
#> [1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8
#>
#> time zone: America/New_York
#> tzcode source: internal
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] bacondecomp_0.1.1 did_2.5.1         wooldridge_1.4-7  plm_2.6-7
#>  [5] stargazer_5.2.3   lfe_3.1.1         Matrix_1.7-4      fixest_0.14.2
#>  [9] ggplot2_4.0.2     dplyr_1.2.0
#>
#> loaded via a namespace (and not attached):
#>  [1] tidyr_1.3.2         sandwich_3.1-1      generics_0.1.4
#>  [4] dreamerr_1.5.0      lattice_0.22-7      bigmemory_4.6.6
#>  [7] digest_0.6.39       magrittr_2.0.4      evaluate_1.0.5
#> [10] grid_4.5.2          RColorBrewer_1.1-3  fastmap_1.2.0
#> [13] jsonlite_2.0.0      Formula_1.2-5       tinytex_0.58
#> [16] purrr_1.2.1         bigmemory.sri_0.1.8 scales_1.4.0
#> [19] stringmagic_1.2.0   numDeriv_2016.8-1.1 Rdpack_2.6.6
#> [22] cli_3.6.6           rlang_1.3.0         rbibutils_2.4.1
#> [25] BMisc_1.4.9         miscTools_0.6-30    withr_3.0.2
#> [28] yaml_2.3.12         otel_0.2.0          DRDID_1.3.0
#> [31] fastglm_0.1.1       tools_4.5.2         parallel_4.5.2
#> [34] uuid_1.2-2          bdsmatrix_1.3-7     maxLik_1.5-2.2
#> [37] vctrs_0.7.1         R6_2.6.1            zoo_1.8-15
#> [40] lifecycle_1.0.5     trust_0.1-9         MASS_7.3-65
#> [43] pkgconfig_2.0.3     pillar_1.11.1       gtable_0.3.6
#> [46] data.table_1.18.2.1 glue_1.8.0          Rcpp_1.1.1
#> [49] collapse_2.1.7      xfun_0.56           tibble_3.3.1
#> [52] lmtest_0.9-40       tidyselect_1.2.1    knitr_1.51
#> [55] farver_2.1.2        xtable_1.8-8        htmltools_0.5.9
#> [58] nlme_3.1-168        labeling_0.4.3      rmarkdown_2.30
#> [61] compiler_4.5.2      S7_0.2.1
```